

Marlon Buchanan

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Education

Simon Fraser University

Bachelor of Applied Science, Sustainable Energy Engineering (Co-op)

Expected Sept 2027

British Columbia Institute of Technology

Diploma, Chemical and Environmental Engineering Technology

May 2021

Work Experience

Mechanical Engineering Intern, FTX Energy Technology – North Vancouver, BC

Sept 2023 – Apr 2024

- Automated fuel cell life-cycle mechanical analysis in MATLAB, generating load profiles used for FEA and shock/vibration compliance testing to meet GB/T standards
- Built a Python web scraper to populate a materials database (75+ entries), accelerating material selection and reducing prototyping research time
- Validated silicone seal prototypes through dimensional and mechanical testing to establish compatibility with PEM fuel cell stack integration

Other Experience

Mechanical Designer, WLF Hydroponic Labs – Surrey, BC

Jan 2025 – Present

- Designed a 3D-printed enclosure in SolidWorks with PCB mounting and cable routing to ensure protection of electronics in a humid hydroponic environment
- Built a hydroponic piping network with 10+ valves and pumps, integrating safety controls to prevent leaks, over-pressure, and system failures
- Fabricated a sheet metal mounting system meeting anti-vibration and lab earthquake safety requirements

Vehicle Dynamics Member, Team Phantom SFU Formula SAE – Surrey, BC

Sept 2023 – May 2025

- Designed and manufactured custom seat brackets accommodating drivers from 152–187 cm and 49–102 kg while meeting FSAE ergonomic and safety standards
- Simulated and optimized harness mounting tabs in ANSYS to achieve a 15 kN pullout load requirement
- Created a safety harness test plan with fixture designs to validate FEA results through physical load testing

Electrical Designer, Women in Clean Tech (WiCT) SFU – Surrey, BC

Jan 2024 – May 2024

- Led a cross-functional team to implement an energy-efficient lighting system compliant with ASHRAE 189.1
- Implemented occupancy and daylight sensor controls to reduce projected lighting energy consumption

Projects

Dynamic Solar Shade with Maximum Power Point Tracking

Sept 2025 – Apr 2026

- Designed a synchronous buck converter 4-layer PCB in Altium with EMI mitigation, component selection, and layout optimization
- Simulated and analyzed gate driver and op-amp IC performance in LTSpice to verify stability
- Designed and hand-soldered 8+ breakout boards (2- and 4-layer) for component validation
- Revised PCB layout across multiple bring-up cycles to improve testability and manufacturability

Skills

Design & Simulation: Altium Designer, LTSpice, PLECS, Simulink, SolidWorks, ANSYS Mechanical

Programming Languages: MATLAB, Python, C, C++

Hardware & Prototyping: Oscilloscope, Multi-meter, Soldering, Function Generator, 3D Printing, Water Jet Cutter, Lathe, Mill